Smart Inquiry Triage Assistant

Automating Customer Journey Analytics

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CURRENT STATE BOTTLENECK

The Operational Challenge

Every day, hundreds of customer inquiries arrive across diverse digital channels—including websites, mobile apps, and vehicle configurators. Today, these inquiries are sorted and routed manually: someone reads each message, determines what it is about, assesses its urgency, and manually assigns it to a team.

01 / INGESTION

The Scale Bottleneck

Hundreds of inquiries flood in from all different digital channels, making it hard to scale manually.

02 / DELAYS

Severe Time Delays

Manual reading and sorting creates massive backlogs and frustrates customers.

03 / ROUTING

Inconsistent Routing

Human errors lead to misdirected tickets and wasted agent effort.

Manual triage is too slow, too inconsistent, and cannot scale with the company's growth.

REAL-TIME INTELLIGENT DISPATCH

The Solution: Instant Retrieval & Routing

An AI-powered Smart Inquiry Triage Assistant automates manual dispatch end-to-end—instantly classifying intent, deriving priority from historical precedents, and providing context-aware resolution notes for human agents.

01 / ROUTING

Instant Classification & Routing

The moment a ticket hits the system, the AI reads it, determines the category, and routes it directly to the correct team.

02 / PRIORITY

Automated Prioritization

By using similarity search over historical cases, the AI instantly flags the new issue as low, medium, or high priority.

03 / ACCELERATION

Agent Acceleration

The AI generates a short 1-2 line resolution note based on past solutions, giving the human agent a massive head start.

INBOUND

User Inbound Query

Raw ticket text or issue description

→

AI CORE

Smart Inquiry Triage System

Vector retrieval & LLM reasoning

→

DISPATCH

Responsible Department Queue

Technical Support / Warranty / Logistics

Taking a manual sorting process that takes hours and shrinking it down to milliseconds.

CORE TECHNOLOGY CHOICES

System Architecture & Tech Stack

🗄️ RAG & Vector Storage

ChromaDB combined with Gemini Embedding models for high-speed similarity search over historical precedents.

🧠 LLM Reasoning Engine

Gemini 1.5 Flash selected for fast intent classification, priority derivation, and structured output generation.

⚙️ Orchestration Framework

LangGraph utilized to build a state-based workflow from ingestion to routing instead of fragile linear scripts.

💻 User Interface

Streamlit wired directly into the backend triage pipeline for real-time interactive testing sessions.

DECISION LOGIC & EXECUTION

Core Processing & Hybrid Reasoning

1. Vector Retrieval (Top-K Context)

Incoming queries are embedded and sent to ChromaDB to pull the most relevant historical precedents and past resolutions.

2. Hybrid Processing (Classification, Priority, Routing & Resolution)

We use a hybrid approach that merges the retrieved Top-K elements with canonical data and sends them to Gemini 1.5 Flash in a single engineered prompt.

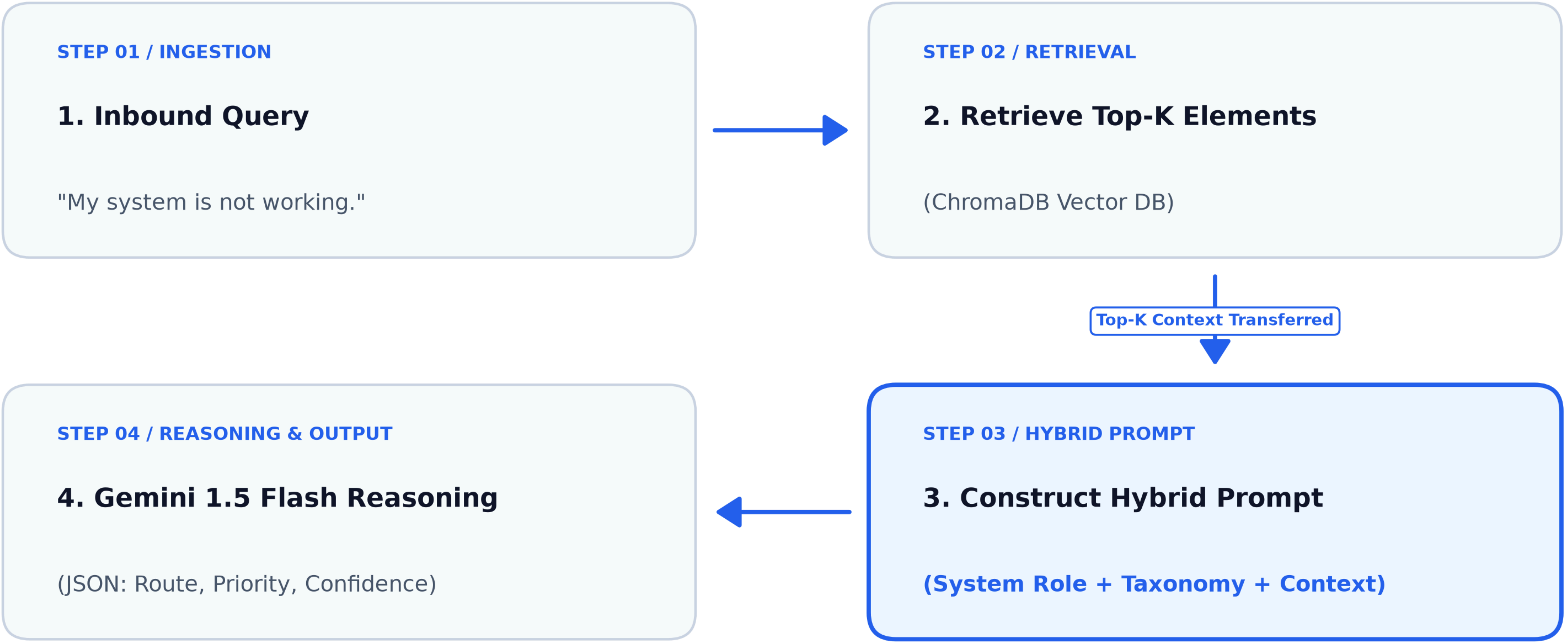
3. Comprehensive Prompt Structure & Calculations

```
Prompt = Role Description + Taxonomy.json + Top-K Retrieved Context + Output Structure + Computation Rules (Priority & Confidence Score)
```

The final prompt integrates canonical business definitions with historical precedents to ensure deterministic, structured LLM reasoning.

STATE-BASED ORCHESTRATION ARCHITECTURE

LangGraph Execution Flow



ENGINEERING & ARCHITECTURAL CHOICES

Key Decisions & Trade-offs

01 / Architectural Approach

Hybrid Approach (LLM + RAG Retrieval) vs. Pure Vector Similarity / KNN. Choosing semantic reasoning over basic spatial clustering for robust intent control.

02 / Advantages of the Hybrid Approach

High explainability and adaptability. Injecting canonical taxonomy content with the LLM allows dynamic updates to business rules without rebuilding vector stores, while easily supporting future human-in-the-loop workflows.

03 / Trade-offs & Costs

Increased round-trip latency and operational token expense. Involving an LLM for final classification adds processing time compared to pure vector space distance calculations.

OPERATOR EXPERIENCE & VALIDATION

Live Demo & Interface

DEMO 01 / WARRANTY TRIAGE

How long is the warranty valid for my new car?

query: How long is the warranty valid for my new car?

category: warranty

priority: low

routed queue: Warranty Claims Team

confidence: 0.98

resolution notes: Provide standard factory warranty duration and mileage limits, or request the customer's VIN to check exact coverage expiration details.

retrieved past cases:

- Can you confirm my warranty end date and mileage limit?
- Can I transfer the remaining warranty to a new owner?
- Does my warranty include roadside assistance?

DEMO 02 / ORDER TRACKING TRIAGE

How can I see my order status?

query: How can I see my order status?

category: ordering

priority: low

routed queue: Order Management Team

confidence: 0.98

resolution notes: Provide instructions on how to track order status through the customer portal or assist directly by looking up account/order details.

retrieved past cases:

- What is the current status of my order placed three weeks ago?
- Can I track my order without logging into the dealer portal?
- Can you confirm the estimated delivery window for order number 88213?

ENTERPRISE EVOLUTION

Future Roadmap & Scaling Strategy

01 / Domain & Taxonomy Expansion

Scaling vector data volumes and expanding the canonical `taxonomy.json` to onboard multi-region teams and new service departments seamlessly.

02 / Enterprise Document RAG & Citations

Ingesting official company wikis, manuals, and policies into a dedicated RAG index so the agent can cite exact official documentation sources.

03 / Rich Resolution Playbooks

Upgrading agent acceleration to retrieve and synthesize step-by-step histories of how complex edge-cases were solved in past instances.

04 / Human-in-the-Loop Active Learning

Logging human agent overrides and edits back into ChromaDB as verified gold-standard examples so the system self-optimizes over time.

QUESTIONS & ANSWERS

Q&A

Open Discussion & Feedback